



EDU TERIA

72nd BPSC

Prelims Online Batch

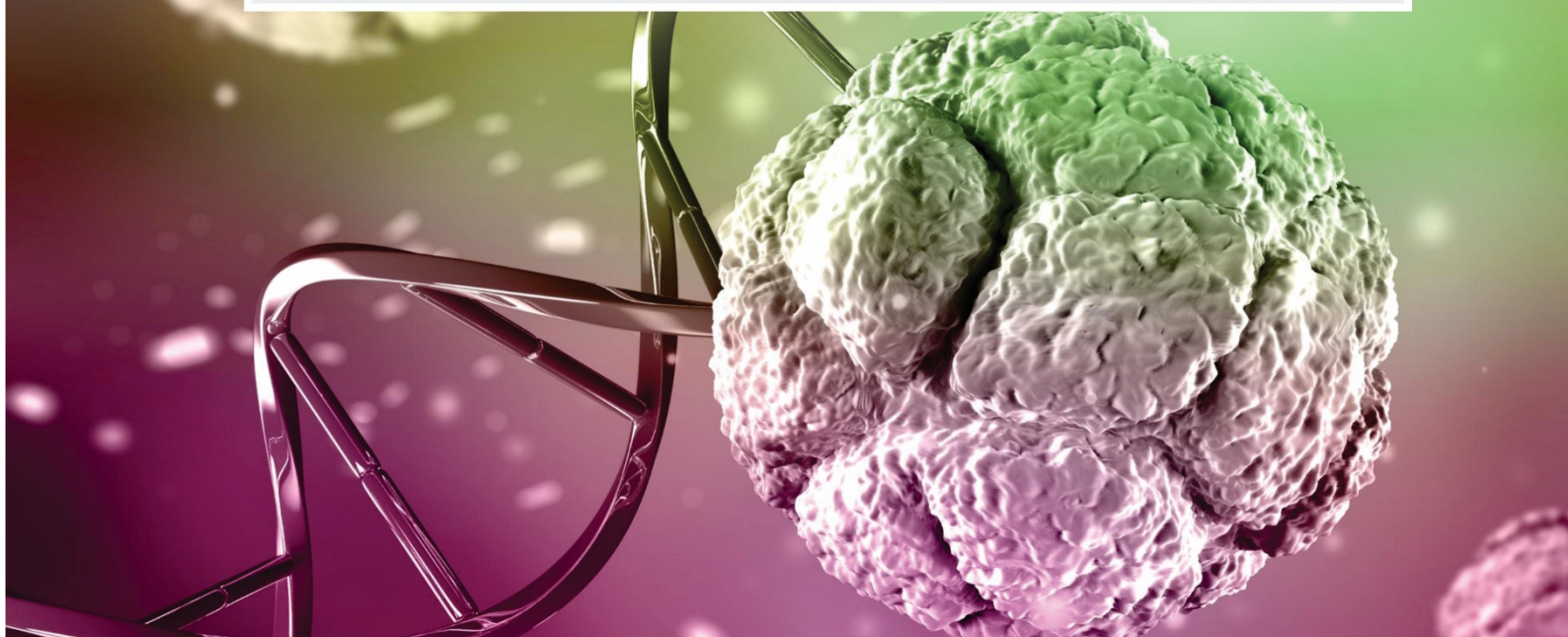
BIOLOGY

Topic wise
Notes

LECTURE

02

Cell Organelles



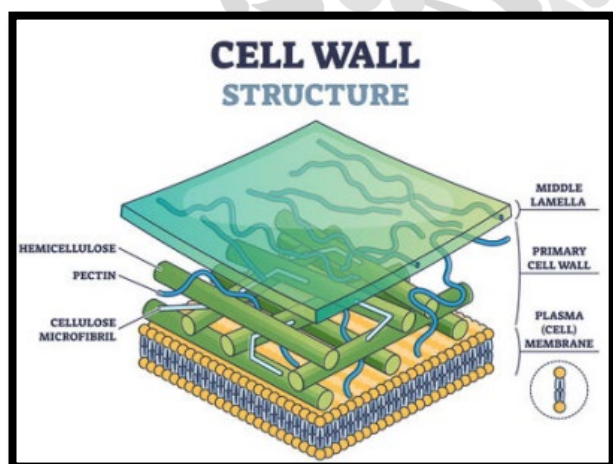
CELL ORGANELLES

Cell organelles are the specialized structures present inside a cell that perform specific functions necessary for the survival, growth, and proper functioning of the cell. Just like different organs in the human body (heart, lungs, brain) perform different tasks, similarly cell organelles act as the internal organs of the cell.

Different types of for Cell Organelles

Cell Wall

- Present outside the cell membrane in plant cells.
- Made up of cellulose.
- Provides rigidity, protection and shape.
- Prevents the cell from bursting due to excess water.



Composition:

- Plants: Composed of cellulose, hemicellulose, pectins, and proteins.
- Algae: Composed of cellulose, galactans, mannans, and minerals like calcium carbonate.
- Fungi: Composed of chitin and glucans.

Structure:

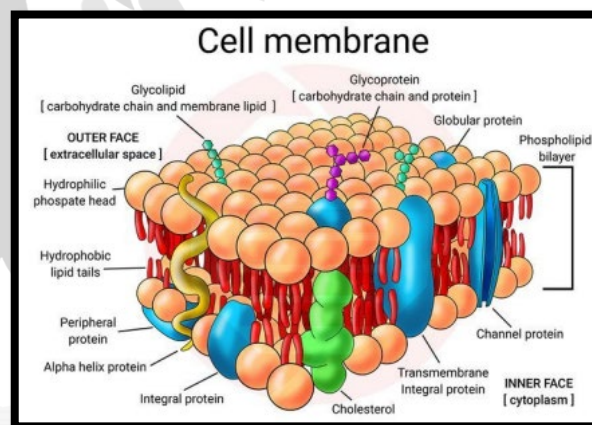
- Middle Lamella: The outermost layer, mainly composed of calcium pectate, which holds/glues different neighboring cells together.
- Primary Wall: The inner wall, capable of growth, which diminishes as the cell matures.
- Secondary Wall: Formed on the inner side (towards the membrane) of the cell as the cell matures.

Functions:

- Support & Protection: Provides structural strength, rigidity, and shape to the plant cell.
- Protection: Shields against mechanical stress, physical shock, and pathogens.

Cell Membrane:

- The cell membrane (plasma membrane) is a thin, flexible, selectively permeable, and living boundary found in all plant and animal cells, separating the cytoplasm from the exterior.
- It is the outermost covering of the cell.
- Made up of lipid bilayer and proteins.
- It is selectively permeable (allows only some substances to pass).
- Controls movement of materials like nutrients, oxygen and waste
- Maintains internal environment of the cell (homeostasis).



Structure - The Fluid Mosaic Model (1972):

- Proposed by Singer and Nicolson
- **Components:** Phospholipid bilayer, proteins, carbohydrates, and cholesterol.
- **Lipid Bilayer:** Phospholipids are arranged with hydrophobic tails facing inward (protected from water) and hydrophilic heads facing outward.



Structure - The Fluid Mosaic Model (1972):

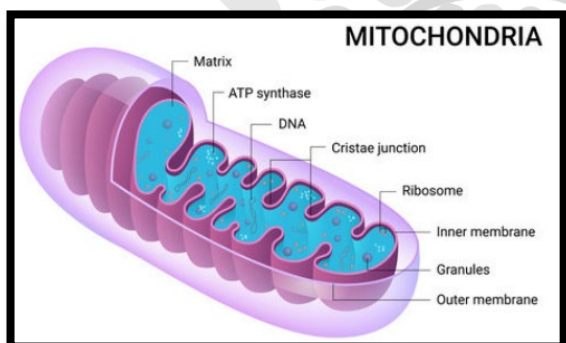
- **Fluidity:** The membrane is fluid, allowing lateral movement of proteins and lipids, which enables functions like cell growth, intercellular junctions, secretion, and endocytosis.

Proteins:

- **Integral/Intrinsic Proteins:** Partially or totally buried in the bilayer.
- **Peripheral/Extrinsic Proteins:** Lie on the surface.
- **Composition Variability:** Human erythrocyte (RBC) membrane has protein and lipids.

Mitochondria

- Mitochondria are double-membrane-bound, sausage-shaped, or cylindrical organelles (0.2-1.0 diameter, 1.0-4.1 length) found in eukaryotic cells, excluding prokaryotes.
- Known as "powerhouses of the cell," they generate ATP through aerobic respiration.
- They are semi-autonomous, containing their own circular DNA, 70S ribosomes, and RNA.



Function of Mitochondria

- Functions of Mitochondria
- Production of ATP (energy currency of the cell)
- Site of aerobic respiration
- Regulation of cell metabolism
- Helps in apoptosis (programmed cell death)
- Oxidation of carbohydrates and fatty acids
- Heat production in some specialized cells.

Structure of Mitochondria

- Mitochondria consist of the following parts:

Outer Membrane

- Smooth and continuous.
- Permeable to small molecules and ions.

- Covers the entire mitochondrion.

Inner Membrane

- Folded inward to form structures called Cristae.
- These folds increase the surface area for energy production.
- Contains enzymes needed for respiration.

Cristae

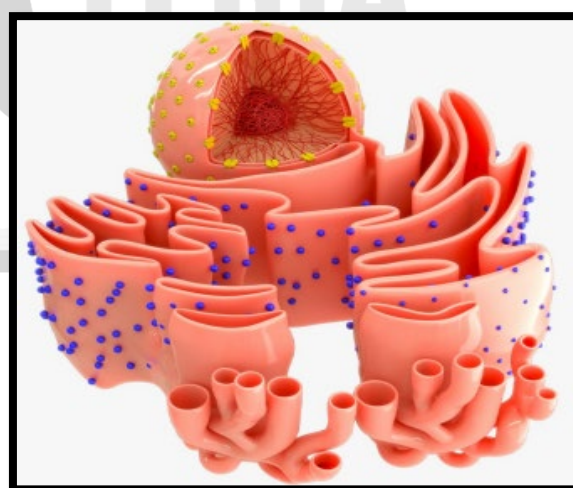
- Finger-like projections of the inner membrane.
- Help in increasing ATP production efficiency.

Matrix

- Inner fluid-filled space.
- Contains:
 - Enzymes
 - Mitochondrial DNA (mtDNA)
 - Ribosomes.

Endoplasmic Reticulum (ER)

- The Endoplasmic Reticulum (ER) is a network of tiny tubular structures scattered in the cytoplasm, dividing the intracellular space into luminal (inside ER) and extra-luminal (cytoplasm) compartments.
- It is continuous with the outer nuclear membrane.
- The Rough ER (RER) has ribosomes for protein synthesis, while the Smooth ER (SER) lacks ribosomes and synthesizes lipids/steroids.



- **Structure:** Network of membrane-bound tubes and sheets (cisternae).
- **Location:** Found in all eukaryotic cells (absent in prokaryotes, RBCs).



**Types:**

- Rough Endoplasmic Reticulum (RER): Ribosomes attached to the surface; prominent in protein-synthesizing cells.
- Smooth Endoplasmic Reticulum (SER): No ribosomes; looks smooth.

Functions of Endoplasmic Reticulum

- **RER:** Protein synthesis and folding.
- **SER:** Synthesis of lipids, steroidal hormones, and detoxification of drugs/poisons.
- **Connectivity:** The ER membrane is often continuous with the outer membrane of the nucleus.
- Nucleus: Robert Brown (1833)
- **Mitochondria:** Richard Altmann (1894, first recognized); Carl Benda (1898, coined term)
- **Golgi Apparatus:** Camillo Golgi (1897/1898)

